

Lilian Linh Tran

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PROFESSIONAL SUMMARY

Software Engineer with an MS in Software Engineering and a strong background in cloud services and modern hosting. Analytical problem-solver skilled at identifying critical system bottlenecks, diagnosing root causes, and designing targeted, high-performance solutions. Experienced in optimizing large-scale data pipelines (2M+ data points) and building production-grade backend architectures.

EDUCATION

San Jose State University, San Jose, CA

Graduated May 2026

Master of Science in Software Engineering

University of California, Berkeley, Berkeley, CA

Graduated May 2023

Bachelor of Arts in Computer Science & Data Science

- **Upsilon Pi Epsilon (UPE):** Member of the honor society for computing and information disciplines
- **Course Staff:** Structure and Interpretation of Computer Programs, Discrete Mathematics and Probability Theory
- **ICPC:** Top 30 Pacific Northwest Region, Rank 8 Northern California

RELEVANT SKILLS

Languages: Python, Java, Go, SQL, Kotlin, TypeScript

Backend/Systems: Data structures, Algorithms, Distributed systems, REST APIs, PostgreSQL, MongoDB, Linux

Testing: pytest, JUnit

Cloud & DevOps: Docker, AWS, GCP, Kubernetes, Terraform, CI/CD, Git, Agile (SDLC)

EXPERIENCE

Software Engineer

Nov. 2023 – Jul. 2024

Saigon Fabrication, San Jose, CA

- Built distributed data processing services using client-server architecture, improving system reliability and accelerating analysis of large datasets by 10%
- Designed and implemented high-throughput REST APIs and backend services using Python and Django
- Developed internal debugging and monitoring tools to diagnose issues across distributed services
- Automated validation workflows and CI/CD pipelines to streamline continuous delivery and improve deployment reliability
- Monitored and troubleshoot production workflows to maintain performance, availability, and data integrity

AI Research Assistant

May. 2023 – Oct. 2023

The University of Texas Health Houston, Houston, TX

- Designed scalable data processing pipelines with parallel computation, improving throughput and efficiency for large-scale datasets
- Developed NLP-based systems for clustering and retrieval across 69K+ medical code entities, improving data accessibility
- Applied dimensionality reduction (PCA, LSA, LDA) to optimize high-dimensional data processing
- Implemented and evaluated ML models and integrated them into backend workflows with reproducible pipelines

AI Engineer Intern

May 2022 – Aug. 2022

TMA Solutions, Ho Chi Minh City, Vietnam

- Devised data preprocessing pipelines for 2M+ data points, improving model precision by 5%
- Built GPU-enabled training environments and CI/CD workflows to support efficient training and deployment

PROJECTS

Tutor Scheduling App (*Next.js, React, TypeScript, PostgreSQL, Prisma, Supabase, NextAuth, Nodemailer*)

- Built and owned a full-stack system with backend services and APIs to support concurrent scheduling workflows
- Designed concurrency-safe database architecture to ensure consistency under concurrent access (PostgreSQL, Prisma, Auth0)

BookTable – Full-Stack Restaurant Reservation Platform (*React, TypeScript, FastAPI, PostgreSQL, Firebase, Google Maps API*)

- Developed and debugged full-stack functionality, resolving issues in search, pricing filters, and review workflows while improving application reliability and user experience.
- Integrated Firebase Cloud Storage for image uploads, Google Maps API for restaurant location services, and email notifications, while creating mock database data to support backend development and testing.

LEADERSHIP & ACTIVITIES

Senator, De Anza Student Government: Managed \$1M+ annual budget supporting campus organizations